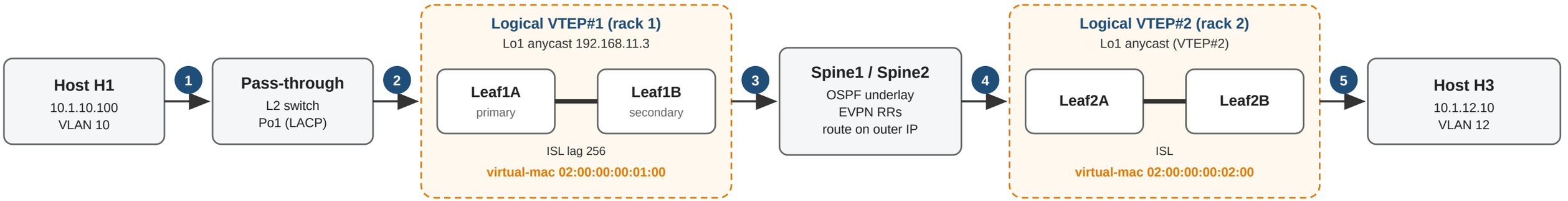
Routed flow: H1 (VLAN 10, rack 1) → H3 (VLAN 12, rack 2) · AOS-CX symmetric IRB with VSX logical VTEPs

Addresses from the AOS-CX 10.16 EVPN-VXLAN guide example (Leaf1A/1B and Leaf2A/2B). Host MACs are placeholders.



1 H1 → pass-through

H1 ARPs for gateway 10.1.10.1 and sends the frame to the anycast gateway MAC.

Eth src	Eth dst	IP	TTL
H1-MAC	12:00:00:00:01:00	10.1.10.100 → 10.1.12.10	64

2 Pass-through → Leaf1A or Leaf1B

dst MAC learned on Po1 → LAG hash (MAC/IP/L4) picks ONE member. Frame unchanged.

Eth src	Eth dst	IP	TTL
H1-MAC	12:00:00:00:01:00	10.1.10.100 → 10.1.12.10	64

3 Leaf1A/1B: route + encapsulate

dst = own gateway MAC → route in VRF1 → /32 from Leaf2 type-2 (host-route) → VTEP#2, L3VNI 100001, Router-MAC 02:00:00:00:02:00.

Outer IP	UDP	VXLAN VNI	Inner src	Inner dst	Inner IP / TTL
192.168.11.3 → VTEP#2 Lo1	dst 4789	100001 (L3VNI)	02:00:00:00:01:00	02:00:00:00:02:00	→ 10.1.12.10 / 63

4 Spines

Route on the OUTER IP only (ECMP to Leaf2A or Leaf2B via the anycast Lo1). Inner packet untouched.

Outer IP	UDP	VXLAN VNI	Inner
192.168.11.3 → VTEP#2 Lo1	dst 4789	100001	unchanged

5 Leaf2A/2B → H3

Decap: VNI 100001 → VRF1; inner dst = own virtual-mac → route into VLAN 12 → ARP entry for H3 → deliver.

Eth src	Eth dst	IP	TTL
12:00:00:00:01:00	H3-MAC	10.1.10.100 → 10.1.12.10	62

LACP (control only) – system-mac

Leaf1A and Leaf1B both send LACPDUs with
Actor System ID = 02:00:00:00:01:00 (vsx system-mac)
→ pass-through sees ONE partner → bundles both links into Po1.
Never appears in data frames.

EVPN control plane (before the flow)

Leaf2A/2B learn H3 and advertise a type-2 MAC+IP via the RRs:
10.1.12.10 / H3-MAC, L2VNI + L3VNI 100001,
RT 65001:1, Router-MAC = 02:00:00:00:02:00
→ Leaf1A/1B install /32 in VRF1 with that router-MAC.

Which MAC is used where

- active-gateway MAC 12:00:00:00:01:00 – hosts ↔ gateway (same on all leaves)
- virtual-mac – inner dst on L3VNI (one per VSX pair)
- system-mac – LACP Actor System ID only
- L3VNI 100001 – routed traffic between racks